



6A. Algebraic Expression

- Write the following using literals, numbers and signs of basic operations.
 - x increased by 12
 - y decreased by 7
 - The difference of a and b. when $a > b$
 - 5 times x added to 7 times y
 - Sum of x and the quotient of y by 5
 - x taken away from 4
 - 2 less than the quotient of x by y
 - x multiplied by itself
 - Twice x increased by y
 - Thrice x added to y squared
 - x cubed less than y cubed
- Write the following in the exponential form:
 - $b \times b \times b \times \dots 15$ times
 - $y \times y \times y \times \dots 20$ times
 - $14 \times a \times a \times a \times a \times b \times b \times b$
 - $6 \times x \times x \times y \times y$
 - $3 \times z \times z \times z \times y \times y \times x$
- Write down the following in the product form:
 - $x^2 y^4$
 - $6y^5$
 - $9xy^2 z$
 - $10a^3 b^3 c^3$
- If $a = 2$ and $b = 3$, find the value of
 - a + b
 - $a^2 + ab$
 - $ab - a^2$
 - $2a - 3b$
 - $5a^2 - 2ab$
 - $a^3 - b^3$
- If $x = 1$, $y = 2$ and $z = 5$, find the value of
 - $3x - 2y + 4z$
 - $x^2 + y^2 + z^2$
 - $2x^2 - 3y^2 + z^2$
 - $xy + yz - zx$
 - $2x^2 y - 5yz + xy^2$
 - $x^3 - y^3 - z^3$
- If $p = -2$, $q = -1$ and $r = 3$, find the value of
 - $p^2 + q^2 - r^2$
 - $2p^2 - q^2 + 3r$
 - $p - q - r$
 - $p^3 + q^3 + r^3 + 3pqr$
 - $p^4 + q^4 - r^4$
 - $3p^2 q + 5pq^2 + 2pqr$
- Write the coefficient of
 - x in $13x$
 - y in $-5y$
 - y^2 in $8xy^2 z$
 - z in $-7xz$
 - p in $-2pqr$
 - a in $6ab$
 - x^3 in x^3
 - x^2 in $-x^2$
 - x^2 in πx^2
- Write the numerical coefficient of
 - ab
 - $-6bc$
 - $7xyz$
 - $-2x^3 y^2 z$
 - $-bc^2$
- Write the constant term of
 - $3x^2 + 5x + 8$
 - $2x^2 - 9$
 - $4y^2 - 5y + \frac{3}{5}$
 - $z^3 - 2z^2 + z - \frac{8}{3}$
- Identify the monomials, binomials and trinomials in the following:
 - $-2xyz$
 - $5 + 7x^3 y^3 z^3$
 - $-5x^3$
 - $a + b - 2c$
 - $xy + yz - zx$
 - x^5
 - $2x + 1$
 - -14
 - $ax^3 + bx^3 + cx + d$
- Write all the terms of the algebraic expressions:
 - $4x^5 - 6y^4 + 7x^2 y - 9$
 - $9x^3 - 5z^4 + 7x^3 y - xyz$
- Identify the like terms in the following:
 - $a^2, b^2, -2a^2, c^2, 4a$
 - $3x, 4xy, -yz, -zy, \frac{1}{2}zy$

- $-2xy^2, x^2 y, 5y^2 x, x^2 z$
- $abc, ab^2 c, acb^2, c^2 ab, b^2 ac, a^2 bc, cab^2$

Add:

- $3x, 7x$
 - $7y, -9y$
 - $2xy, 5xy, -xy$
 - $3x, 2y$
 - $2x^2, 3x^2, 7x^2$
 - $6a^3, -4a^3, 10a^3, -8a^3$
 - $7xyz, -5xyz, 9xyz, -8xyz$
 - $x^2 - a^2, -5x^2 + 2a^2, -4x^2 + 4a^2$
- $x - 3y - 2z, 5x + 7y - z$ and $-7x - 2y + 4z$
 - $m^2 - 4m + 5, -2m^2 + 6m - 6$ and $-m^2 - 2m - 7$
 - $2x^2 - 3xy + y^2, -7x^2 - 5xy - 2y^2$ and $4x^2 + xy - 6y^2$
 - $4xy - 5yz - 7zx, -5xy + 2yz + zx$ and $-2xy - 3yz + 3zx$
- $3a - 2b + 5c, 2a + 5b - 7c, -a - b + c$
 - $8a - 6ab + 5b, 6a - ab - 8b, -4a + 2ab + 3b$
 - $2x^3 - 3x^2 + 7x - 8, -5x^3 + 2x^2 - 4x + 1, 3 - 6x + 5x^2 - x^3$
 - $2x^2 - 8xy + 7y^2 - 8xy^2, 2xy^2 + 6xy - y^2 + 3x^2, 4y^2 - xy - x^2 + xy^2$
 - $x^3 + y^3 - z^3 + 3xyz, -x^3 + y^3 + z^3 - 6xyz, x^3 - y^3 - z$
 - $2 + x - x^2 + 6x^3, -6 - 2x + 4x^2 - 3x^3, 2 + x^2, 3 - x^3 + 4x - 2x$

Subtract:

- 5x from 2x
 - xy from 6xy
 - 3a from 5b
 - 7x from 9y
 - $10x^2$ from $-7x^2$
 - $a^2 - b^2$ from $b^2 - a^2$
- $5a + 7b - 2c$ from $3a - 7b + 4c$
 - $a - 2b - 3c$ from $-2a + 5b - 4c$
 - $5x^2 - 3xy + y^2$ from $7x^2 - 2xy - 4y^2$
 - $6x^3 - 7x^2 + 5x - 3$ from $4 - 5x + 6x^2 - 8x^3$
 - $x^3 + 2x^2 y + 6xy^2 - y^3$ from $y^3 - 3xy^2$
 - $-11x^2 y^2 + 7xy - 6$ from $9x^2 y^2 - 6xy + 9$
 - $-2a + b + 6d$ from $5a - 2b - 3c$

Simplify:

- $2p^3 - 3p^2 + 4p^5 - 6p^3 + 2p^2 - 8p - 2 + 6p + 8$
 - $2x^2 - xy + 6x - 4y + 5xy - 4x + 6x + 3y$
 - $x^4 - 6x^3 + 2x - 7 + 7x^3 - x + 5x^2 + 2 - x^4$
- $a - (b - 2a)$
 - $4x - (3y - x + 2z)$
- $(a^2 + b^2 + 2ab) - (a^2 + b^2 - 2ab)$
 - $-3(a + b) + 4(2a - 3b) - (2a - b)$
 - $-4x^2 + \{(2x^2 - 3) - (4 - 3x^2)\}$
 - $-2(x^2 - y^2 + xy) - 3(x^2 + y^2 - xy)$
 - $a - [2b - \{3a - (2b - 3c)\}]$
 - $-x + [5y - \{x - (5y - 2x)\}]$
 - $86 - [15x - 7(6x - 9) - 2\{10x - 5(2 - 3x)\}]$
 - $12x - [3x^2 + 5x^2 - \{7x^2 - (4 - 3x - x^3) + 6x^3\} - 3x]$
 - $5a - [a^2 - \{2a(1 - a + 4a^2) - 3a(a^2 - 5a^3)\}] - 8a$
 - $3 - [x - \{2y - (5x + y - 3) + 2x^2\} - (x^2 - 3y)]$
 - $xy - [yz - zx - \{yx - (3y - xz) - (xy - zy)\}]$
 - $2a - 3b - [3a - 2b - \{a - c - (a - 2b)\}]$
 - $-a - [a + \{a + b - 2a - (a - 2b)\} - b]$

6B. Simple Equations

1. Write each of the following statements as an equation:

- (a) 5 times a number equals 40.
 (b) A number Increased by 8 equals 15.
 (c) 25 exceeds a number by 7.

2. Write a statement for each of the equations, given below:

- (a) $x - 7 = 14$ (b) $2y = 18$ (c) $11 + 3x = 17$
 (d) $2x - 3 = 13$ (e) $12y - 30 = 6$ (f) $\frac{2z}{3} = 8$

3. Verify by substitution that

- (a) the root of $3x - 5 = 7$ is $x = 4$
 (b) the root of $3 + 2x = 9$ is $x = 3$

4. Solve each of the following equations by the trial-and-error method:

- (a) $y + 9 = 13$ (b) $x - 7 = 10$ (c) $4x = 28$
 (d) $3y = 36$ (e) $11 + x = 19$ (f) $\frac{x}{3} = 4$

Solve each of the following equations and. verify the answer in each case:

5. (a) $x - 1 = 0$ (b) $x + 1 = 0$ (c) $x - 1 = 5$
 (d) $x + 6 = 2$ (e) $y - 4 = -7$ (f) $y - 4 = 4$
 (g) $y + 4 = 4$ (h) $y + 4 = -4$

6. (a) $3l = 42$ (b) $\frac{b}{2} = 6$ (c) $\frac{p}{7} = 4$
 (d) $4x = 25$ (e) $8y = 36$ (f) $\frac{z}{3} = \frac{5}{4}$
 (g) $\frac{a}{5} = \frac{7}{15}$ (h) $20t = -10$

7. (a) $3n - 2 = 46$ (b) $5m + 7 = 17$
 (c) $\frac{20p}{3} = 40$ (d) $\frac{3p}{10} = 6$

8. (a) $10p = 100$ (b) $10p + 10 = 100$ (c) $\frac{p}{4} = 5$
 (d) $\frac{P}{3} = 5$ (e) $\frac{3p}{4} = 6$ (f) $3s = -9$

- (g) $3s + 12 = 0$ (h) $3s = 0$ (i) $2q = 6$
 (j) $2q - 6 = 0$ (k) $2q + 6 = 0$ (l) $2q + 6 = 12$

9. (a) $2y + \frac{5}{2} = \frac{37}{2}$ (b) $5t + 28 = 10$ (c) $\frac{a}{5} + 3 = 2$
 (d) $\frac{q}{4} + 7 = 5$ (e) $\frac{5}{2}x = 10$ (f) $\frac{5}{2}x = \frac{52}{4}$
 (g) $7m + \frac{10}{2} = 13$ (h) $6z + 10 = -2$ (i) $\frac{3l}{2} = \frac{2}{3}$

- (j) $\frac{2b}{3} - 5 = 3$

10. (a) $2(x + 4) = 12$ (b) $3(n - 5) = 21$ (c) $3(n - 5) = -21$
 (d) $-4(2 + x) = 8$ (e) $4(2 - x) = 8$

11. (a) $4 = 5(p - 2)$ (b) $-4 = 5(p - 2)$
 (c) $16 = 4 + 3(t + 2)$ (d) $4 + 5(p - 1) = 34$
 (e) $0 = 16 + 4(m - 6)$

6C. Exponents and Powers

1. Find the value of:

- (a) 2^6 (b) 9^3 (c) 11^2 (d) 5^4 (e) 6^7 (f) 12^3

2. Express the following in exponential form:

- (a) $6 \times 6 \times 6 \times 6$ (b) $t \times t$
 (c) $b \times b \times b \times b$ (d) $5 \times 5 \times 7 \times 7 \times 7$
 (e) $2 \times 2 \times a \times a$ (f) $a \times a \times a \times c \times c \times c \times c \times d$

3. Express each of the following numbers using exponential notation:

- (a) 512 (b) 343 (c) 729 (d) 3125 (e) 625

4. Identify the greater number, wherever possible, in each of the following?

- (a) 4^3 या 3^4 (b) 5^3 या 3^5
 (c) 2^8 या 8^2 (d) 2^3 या 2^2

5. Express each of the following as product of powers

of their prime factors:

- (a) 648 (b) 405 (c) 540 (d) 3600 (e) 7200

Simplify:

6. (a) 2×10^3 (b) 72×2^2 (c) $2^3 \times 5$
 (d) 3×4^4 (e) 0×10^2 (f) $5^2 \times 3^3$
 (g) $2^4 \times 3^2$ (h) $3^2 \times 10^4$

7. Simplify:

- (a) $(-4)^3$ (b) $(-3)^3 \times (-2)^3$ (c) $(-3)^2 \times (-5)^2$

Using laws of exponents, simplify and write the answer in exponential form:

8. (a) $2^5 \times 2^3$ (b) $p^3 \times p^2$
 (c) $4^3 \times 4^2$ (d) $a^3 \times a^2 \times a^7$
 (e) $5^3 \times 5^7 \times 5^{12}$ (f) $(-4)^{10} \times (-4)^{20}$

9. (a) $2^9 \div 2^3$ (b) $10^8 \div 10^4$
 (c) $9^{11} \div 9^7$ (d) $20^{15} \div 20^{13}$
 (e) $7^{13} \div 7^{10}$ (f) $11^6 \div 11^2$

10. (a) $(6^2)^4$ (b) $(2^2)^{100}$ (c) $(7^{50})^2$ (d) $(5^3)^7$

11. (a) $4^3 \times 2^3$ (b) $2^5 \times b^5$
 (c) $a^2 \times t^2$ (d) $5^6 \times (-2)^6$
 (e) $(-2)^4 \times (-3)^4$ (f) $a^m \times b^m$

12. (a) $4^5 \div 3^5$ (b) $2^5 \div b^5$
 (c) $(-2)^5 \div b^3$ (d) $5^6 \div (-2)^6$

13. (a) 8^0 (b) $(-3)^0$
 (c) $4^0 + 5^0$ (d) $6^0 \times 7^0$

14. (a) $(4)^{-1}$ (b) $(-6)^{-1}$ (c) $(\frac{1}{3})^{-1}$ (d) $(\frac{-2}{3})^{-1}$

15. (a) $3^2 \times 3^4 \times 3^8$ (b) $6^{15} \div 6^{10}$
 (c) $a^3 \times a^2$ (d) $7^x \times 7^2$
 (e) $(5^2)^3 \div 5^3$ (f) $2^5 \times 5^5$
 (g) $a^4 \times b^4$ (h) $(3^4)^3$
 (i) $(2^{20} \div 2^{15}) \times 2^3$ (j) $8^t \div 8^2$

16. Express each of the following as a product of prime factors only in exponential form:
 (a) 108×192 (b) 270 (c) 729×64 (d) 768

17. Simplify:

- (a) $\frac{(2^5)^2 \times 7^3}{8^3 \times 7}$ (b) $\frac{25 \times 5^2 \times t^8}{10^3 \times t^4}$ (c) $\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5}$

18. Write the following numbers in the expanded forms:

279404, 3006194, 2806196, 120719, 20068

19. Find the number from each of the following expanded forms:

- (a) $8 \times 10^4 + 6 \times 10^3 + 0 \times 10^2 + 4 \times 10^1 + 5 \times 10^0$
 (b) $4 \times 10^5 + 5 \times 10^3 + 3 \times 10^2 + 2 \times 10^0$
 (c) $3 \times 10^4 + 7 \times 10^2 + 5 \times 10^0$
 (d) $9 \times 10^5 + 2 \times 10^2 + 3 \times 10^1$

20. Express the following numbers in standard form:

- (a) 5,00,00,000 (b) 70,00,000
 (c) 3,18,65,00,000 (d) 3,90,878
 (e) 39087.8 (f) 3908.78

21. Express each of the following numbers in standard form:

- (a) Diameter of Earth = 12756000 m.
 (b) Distance between Earth and Moon = 384000000 m.
 (c) Population of India in March 2001 = 1027000000.
 (d) Number of stars in a galaxy = 100000000000.
 (e) The present age of universe = 12000000000 years.